

Lab 7 STA 360

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Agenda

In this lab, we will deriving conditional distributions, code a Gibbs sampler, and analyze the output of the Gibbs sampler.

Consider the following Exponential model for observation(s) $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$.¹:

$$p(x|a, b) = ab \exp(-abx)I(x > 0),$$

where the x_i are assumed to be iid for $i = 1, \dots, n$. and suppose the prior is

$$p(a, b) = \exp(-a - b)I(a, b > 0).$$

You want to sample from the posterior $p(a, b|x_{1:n})$. You may assume that

$$a_0 = 0.25, b_0 = 0.25$$

when coding up your Gibbs sampler.

1. Find the conditional distributions needed for implementing a Gibbs sampler.
2. Code up your own Gibbs sampler using part (1).
3. Run the Gibbs sampler, providing convergence diagnostics.
4. Plot a histogram or a density estimate of the estimated posterior using (2) and (3).
5. How do you know that your estimated posterior in (3) is reliable? **If your traceplot for example or running average plot is not "flattening out," then try running more Gibbs iterations.**

Task 1

Consider the following Exponential model for observation(s) $x = (x_1, \dots, x_n)$.²:

$$p(x|a, b) = ab \exp(-abx)I(x > 0)$$

and suppose the prior is

$$p(a, b) = \exp(-a - b)I(a, b > 0).$$

You want to sample from the posterior $p(a, b|x)$.

It is easy to show that the posterior distribution is intractable, hence, we derive the conditional distributions:

$$\begin{aligned} p(\mathbf{x}|a, b) &= \prod_{i=1}^n p(x_i|a, b) \\ &= \prod_{i=1}^n ab \exp(-abx_i) \\ &= (ab)^n \exp\left(-ab \sum_{i=1}^n x_i\right). \end{aligned}$$

¹Please note that the observed data can be found in data-exponential.csv.

²The observed data can be found in data-exponential.csv.

The function is symmetric for a and b , so we only need to derive $p(a|\mathbf{x}, b)$.

This conditional distribution satisfies

$$\begin{aligned}
 p(a|\mathbf{x}, b) &\propto_a p(a, b, \mathbf{x}) \\
 &= p(\mathbf{x}|a, b)p(a, b) \\
 &= (ab)^n \exp\left(-ab \sum_{i=1}^n x_i\right) \times \exp(-a-b)I(a, b > 0) \\
 &\propto_a p(x, a, b) \propto_a a^n \exp(-abn\bar{x} - a)\mathbb{1}(a > 0) = a^{n+1-1} \exp(-(bn\bar{x} + 1)a)\mathbb{1}(a > 0) \propto_a \text{Gamma}(a \mid n + 1, bn\bar{x} + 1).
 \end{aligned}$$

Therefore, $p(a|b, \mathbf{x}) = \text{Gamma}(a \mid n + 1, bn\bar{x} + 1)$ and by symmetry, $p(b|a, \mathbf{x}) = \text{Gamma}(b \mid n + 1, an\bar{x} + 1)$.

Task 2

We now provide code to implement the Gibbs sampler.

```
knitr::opts_chunk$set(cache=TRUE)
library(MASS)
data <- read.csv("data-exponential.csv", header = FALSE)
#####
# This function is a Gibbs sampler
#
# Args
#   start.a: initial value for a
#   start.b: initial value for b
#   n.sims: number of iterations to run
#   data: observed data, should be in a
#         # data frame with one column
#
# Returns:
#   A two column matrix with samples
#     # for a in first column and
#   samples for b in second column
#####
```

```
knitr::opts_chunk$set(cache=TRUE)
sampleGibbs <- function(start.a, start.b, n.sims, data){
  # get sum, which is sufficient statistic. note: sum(x) = n*x_bar.
  x_sum <- sum(data)
  # get n
  n <- nrow(data)
  # create empty matrix, allocate memory for efficiency
  res <- matrix(NA, nrow = n.sims, ncol = 2)
  res[1,] <- c(start.a, start.b)
  for (i in 2:n.sims){
    # sample the values
    res[i,1] <- rgamma(1, shape = n+1,
                      rate = res[i-1,2]*x_sum+1)
    res[i,2] <- rgamma(1, shape = n+1,
                      rate = res[i,1]*x_sum+1)
  }
}
```

```
    return(res)
}
```

Task 3

We now run the Gibbs sampler and produce some results. Finish the rest of this for homework.

```
# run Gibbs sampler
n.sims <- 10000
res <- sampleGibbs(.25,.25,n.sims,data)
head(res)
```

```
##           [,1]      [,2]
## [1,] 0.250000 0.2500000
## [2,] 2.395365 0.1661625
## [3,] 2.649127 0.2017039
## [4,] 2.183022 0.2326039
## [5,] 1.492581 0.3251624
## [6,] 0.950689 0.6800621
```

```
dim(res)
```

```
## [1] 10000      2
```

```
res[1,1]
```

```
## [1] 0.25
```

Task 4 (Finish for homework)

Task 5 (Finish for homework)